Advanced Data Analytics

Accredited by

Data Analytics

- Process of exploring and analyzing data to gain insights and drive informed decision-making within organizations.
- Encompasses a wide range of techniques, methodologies, and tools aimed at leveraging data to improve business performance, enhance operational efficiency, and achieve strategic objectives.

Different Stages in Data Analytics

Collecting the data

- Collection of relevant data from various sources, including internal databases, external sources, and third-party data providers.

Cleaning and preparing the data

- Clean, integrate, and transform the data into a format suitable for analysis.

Exploring and summarizing data

- Visually explore and summarize data to identify patterns, trends, and relationships.

Applying methods and models

- Apply statistical methods and models to quantify relationships between variables, test hypotheses, and make predictions.

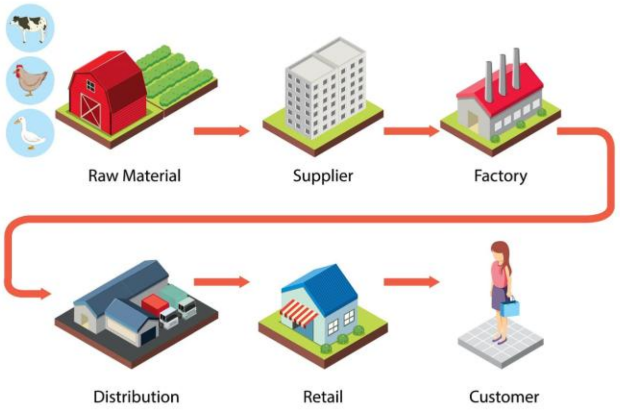
Forecasting/predicting future outcomes

- Forecast future outcomes or trends based on historical data and statistical algorithms

Datamites

Application of Business Analytics

- Customer Segmentation
- Sales Forecasting
- Supply chain optimization
- Risk Analysis
- Operational Efficiency
- Predictive Maintenance



Datamites

Data Sources

- Internal Data. Eg. Customer Data, Sales data.
- External Data. Eg. Social Media Platform, Market research firms, and Government agencies.
- Public Data. Eg. Census data, Weather data.
- Third-party Data: This refers to data that is purchased or acquired from third-party sources, such as data brokers or data providers.

The choice of data sources will depend on the specific objectives of the analysis and the type of insights that are needed.

Data Sources

- Database
- Spreadsheets
- Big Data Platforms
- APIs

Challenges

- Poor data quality, including inaccuracies, inconsistencies, and incompleteness.

So we need accurate data to get accurate results

- Integrating data from disparate sources with different formats and structures presents additional challenges.

So we need a structured format to smooth analysis

- Protecting sensitive data from unauthorized access, breaches, and compliance violations.

So the data should be protected

- Challenges in recruiting and retaining qualified talent with diverse skills like programming, data analysis, statistical modelling, and domain expertise.

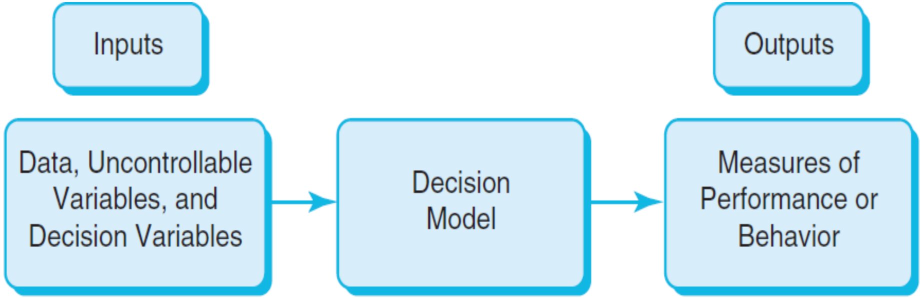
So skill gap is there, proper knowledge is required

- Rapidly evolving landscape of analytics tools and technologies can be overwhelming for organizations to navigate and implement effectively.

So continuous learning mindset is needed.

Building and Optimizing Models

Decision Modeling



- A logical or mathematical representation of a problem or business situation that can be used to understand, analyze, or facilitate making a decision.

- Inputs:
- Uncontrollable variables, which are quantities that can change but cannot be directly controlled by the decision maker.
- Decision variables, which are controllable and can be selected at the discretion of the decision maker.

Example: Inventory Planning

Scenario: A shop owner is planning stock for next week.

Question: How many units should be stocked?

- **Goal**
- Meet customer demand
- Avoid extra stock (waste, storage cost)
- **Stock enough, but not too much**

Step 1: Estimate Demand

Use past data:

- Last week sales = 90 units
- Average weekly sales = 100 units

So, expected demand ≈ **100 units**

Step 2: Understand Costs

If extra stock remains:

- Storage cost
- Risk of damage/spoilage
- This is called: **Holding Cost**

Step 3: Test Different Decisions (Scenarios)

Scenario A: Stock = 200 units

Demand = 100 units

Result:

- 100 units unsold
- Storage cost increases
- Waste/spoilage possible
- **Profit decreases**

Scenario B: Stock = 60 units

Demand = 100 units

Result:

- Not enough products
- Customers unhappy
- Lost sales
- **Profit decreases**

Final Decision (Optimal Point)

We need a **balance point**

- **Best Stock: 100 – 110 units**

Prescriptive Model Pricing

Predictive Analytics with Regression Techniques

Prescriptive Model Pricing

- A firm wishes to determine the best pricing for one of its products in order to maximize profit.
- Analysts determined the following model: Sales = -2.9(price) + 3000
- Total revenue = (price)(sales)
- Fixed cost = 5000
- Total Cost = 10(Sales) + 5000

- Identify the price that maximizes profit, subject to any constraints that might exist.

Sales = -2.9 (Price) + 3000
Total revenue = (Price)(Sales)
Cost = 10(Sales) + 5000

Price	$ 100
Sales	2950
Total Cost	$ 34,500
Cost per unit	$ 12
Profit	$ 260,500

Sales Promotion Decision

Predictive Analytics with Regression Techniques

Sales Promotion Decision

- In the grocery industry, managers typically need to know how best to use pricing, coupons and advertising strategies to influence sales.
- Controlled experiments:

  Grocers implement different combinations of pricing, coupons, and advertising to observe the resulting sales. Analytics helps develop a predictive model of sales as a function of these decision strategies.

Sales Prediction with Multiple Regression Analysis

- Find the pricing and Ads Budget to maximize the Sales
- Use Multiple Linear Regression
- Cost price of the Product is $8
- Minimum Profit desired is $10,000

Cost Price per unit	$9.00
Minimum Profit	$4,000.00
Maximum Ads Budget	$10,000.00
Minimum Ads Budget	$1,000.00

	Coefficients
Intercept	3092.883488
Price	-142.5484493
Ads Budget	0.10225716

Results from Regression Analysis

Sales Promotion Data			
Week	Price	Ads Budget (Thousands USD)	Sales
1	$15.0	1500	1240
2	$16.4	4000	950
3	$13.0	2000	1580
4	$16.5	3000	1190
5	$12.0	1000	1430
6	$11.0	1000	1560
7	$14.3	1000	1060
8	$18.2	3465	800
9	$9.6	1000	1895
10	$15.4	4000	1475
11	$12.0	2000	1560
12	$14.0	1000	1235
13	$16.0	3000	1065
14	$15.0	1000	1025
15	$11.0	2000	1645

Solution: Sales = -142 x Price + 0.10 x Ad+3091

Predictive Analytics with less uncertainty

Stable, well understood systems with established patterns can utilise low-uncertainty predictions effectively.

What is Uncertainty?

Uncertainty means: Lack of clarity , Difficult to predict , System is complex

Key Idea

Predictive analytics works better when: The system is **stable and predictable**

Important Points

- Stable systems give **accurate results**
- Prediction accuracy depends on how **stable and predictable** the system is
- Some systems behave in a **regular and consistent way**
- Other systems may change **suddenly and unpredictably**
-

Example

A grocery store that sells similar quantities every week

- Sales are consistent
- Patterns repeat

Predictive Analytics with Less Uncertainty Means

- Systems behave **consistently**
- Patterns **repeat over time**
- Relationships between variables are **well understood**

KERC Inc – Car Manufacturer

Optimization Model -1

Predictive Analytics with Less uncertainty

KERC Inc – Car Manufacturer

- Has two main car models
  - KERC 501
  - KERC 601

Below table provides the Activity times (hrs) of the models

	Activity time in Hours		
Car Model	Assembly (All parts)	Frame Fitting and Painting	Quality Assurance
KERC 501	8	4	2
KERC 601	10	3	1.6

Datamites

KERC Inc – Car Manufacturer…

KERC Manufacturing Unit has a limited resources for activities as below

Manufacturing Activity	Time Available Hours per Week
Assembly (All parts)	10800
Frame Fitting and Painting	4500
Quality Assurance	2500

Car Model Profitability Table

Car Model	Profit (USD Thousands)
KERK 501	4.9
KERK 601	5.8

Optimize KERC production

Define Objective Function (profit) and find the Optimal manufacturing quantities for car models to maximize the objective function?

WonderZon Retailer

Optimization Model of Logistics to minimize cost

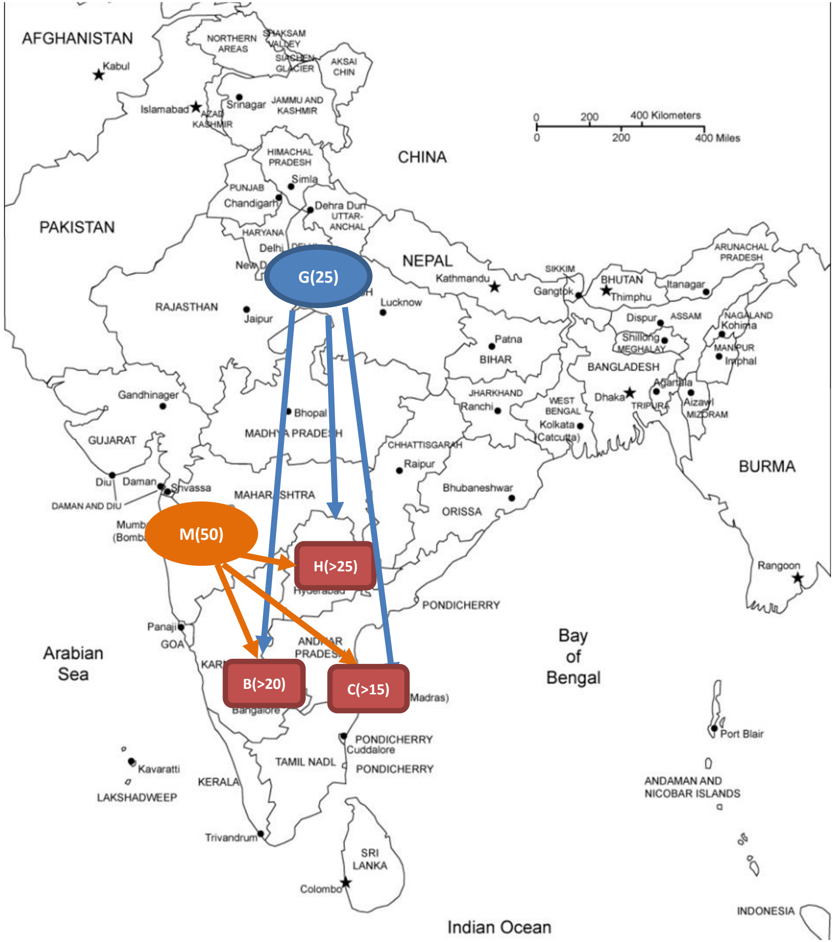
Predictive Analytics with Less uncertainty

Find Objective function for cost and optimize the network for min logistics cost for WonderZon Retailer

Warehouses	Shipment Quantity
Gurgaon	25
Mumbai	50

Delivery Centers	Minimum Quantity
Bengaluru (B)	20
Hyderabad(H)	25
Chennai(C)	15

Logistics Cost in INR per Package		Delivery Centers		
	From / To	**Bengaluru (B)**	**Hyderabad (H)**	**Chennai (C)**
Warehouses	**Gurgaon (G)**	**116**	**106**	**115**
	Mumbai (M)	**110**	**95**	**112**



Predictive Analytics

with High Uncertainty

Internet Usage - Plan Decision

Monte Carlo Simulation

Predictive Analytics with high uncertainty

SMART IT Inc. Internet Plan

SMART IT Inc has been using FLEXI corporate internet plan for a year and thinking about deciding to switch to FAST1000. SMART IT Inc business is seasonal and internet usage varies as per the demand.

Generally bill usage vary from around 1000 GB. Past 6 months usage in GB are 620,1050,1200,1480,870,1600

Find the best plan for SMART IT (Use Monte Carlo Simulation with Normally distribution data)

Corporate Internet Plans	
FLEXI	Flat plan charging $2 per GB
FAST1000	1000 GB included , Post 1000 GB , $1500 + $3.5 per GB

Monte Carlo Simulation

It is a probability model used to predict the probability of various outcomes.

It is used to mitigate the possibility of unpredicted outcomes.

Kickathlon Sport Retailer

Decision Tree & Monte Carlo Simulation

Predictive Analytics with High uncertainty

Kickathlon – Supplier Decision Making

- Kickathlon, A French based Sports Retailer, plans to introduce a new product, TrekBareFoot – A tough Polyester based Socks, which can protect the foot while trekking barefoot.
- TrekBareFoot is scheduled to launch in next Summer beginning so should be manufactured well in advance.
- The price tag for TrekBareFoot is fixed at **$200/-**
- If the marketing efforts create a strong demand then about **15000** units can be sold, on the lower side, weak demand might end up with only **6000** units.
- There is a 50:50% chance for strong and weak demand in the market.
- TrekBareFoot being a specialized product, Kickathlon found only two suppliers one in USA and other in China. The pricing is as below.

	USA (A)	China (C)
Supplier Capacity	**6000**	**15000**
Fixed Contract Charges	**0**	**$500,000**
Unit Product cost	**$150**	**$130**
Profit per unit	**$50**	**$70**

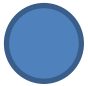
Decision Trees

A Decision Tree is a visual and analytical decision support tool that helps map out decisions and their possible consequences, including risks, costs and benefits.

Decision Tree – Key Components

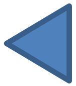
Decision Nodes

There are decision points where it branches to more than one decisions



Chance / Event Nodes

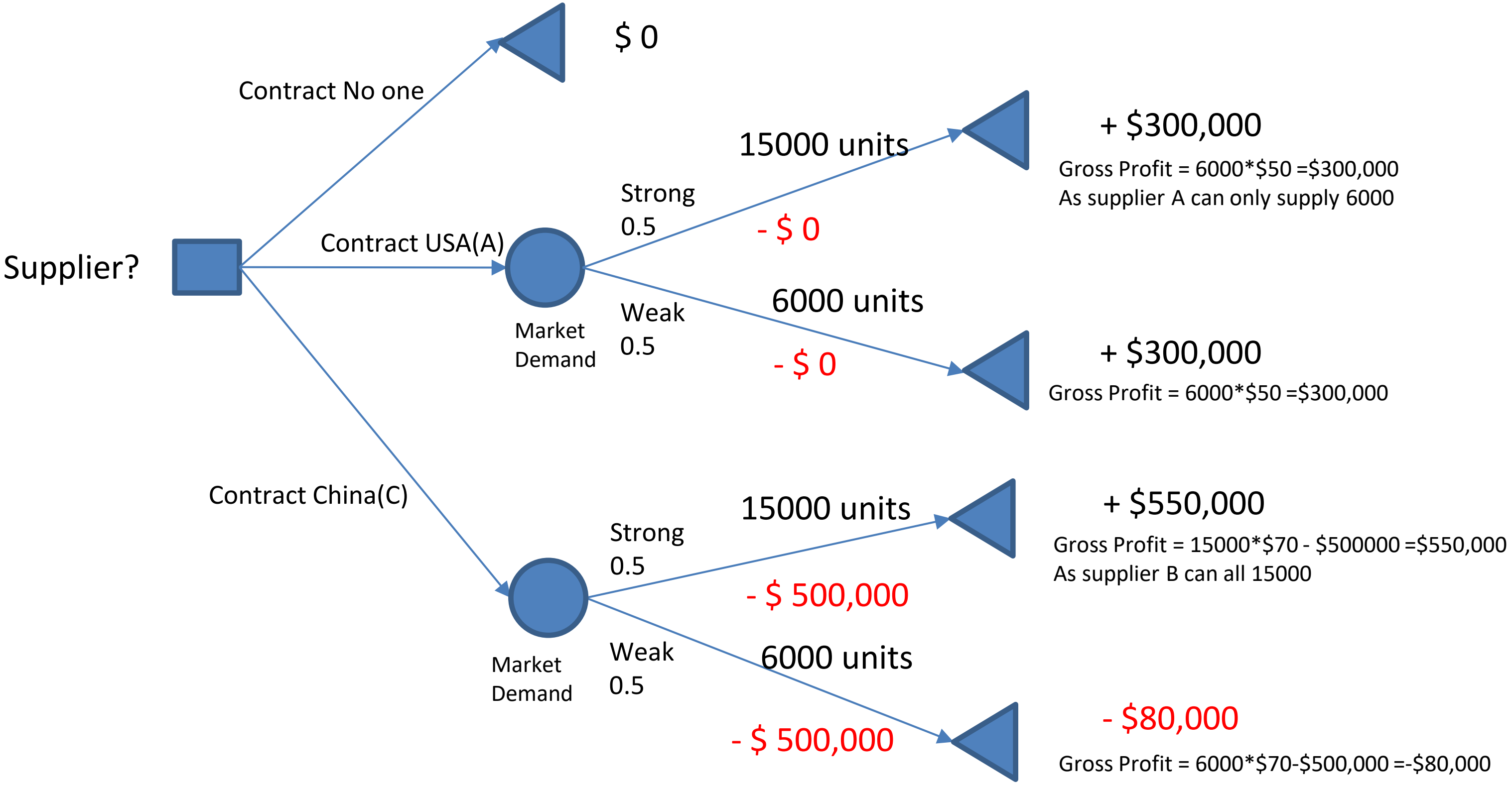
There are points of uncertainty, which is defined by probability / random variable



Outcome Nodes

There are the outcome variable (Objective function), which needs to be either maximized or minimized.

Kickathlon – Supplier Decision Tree



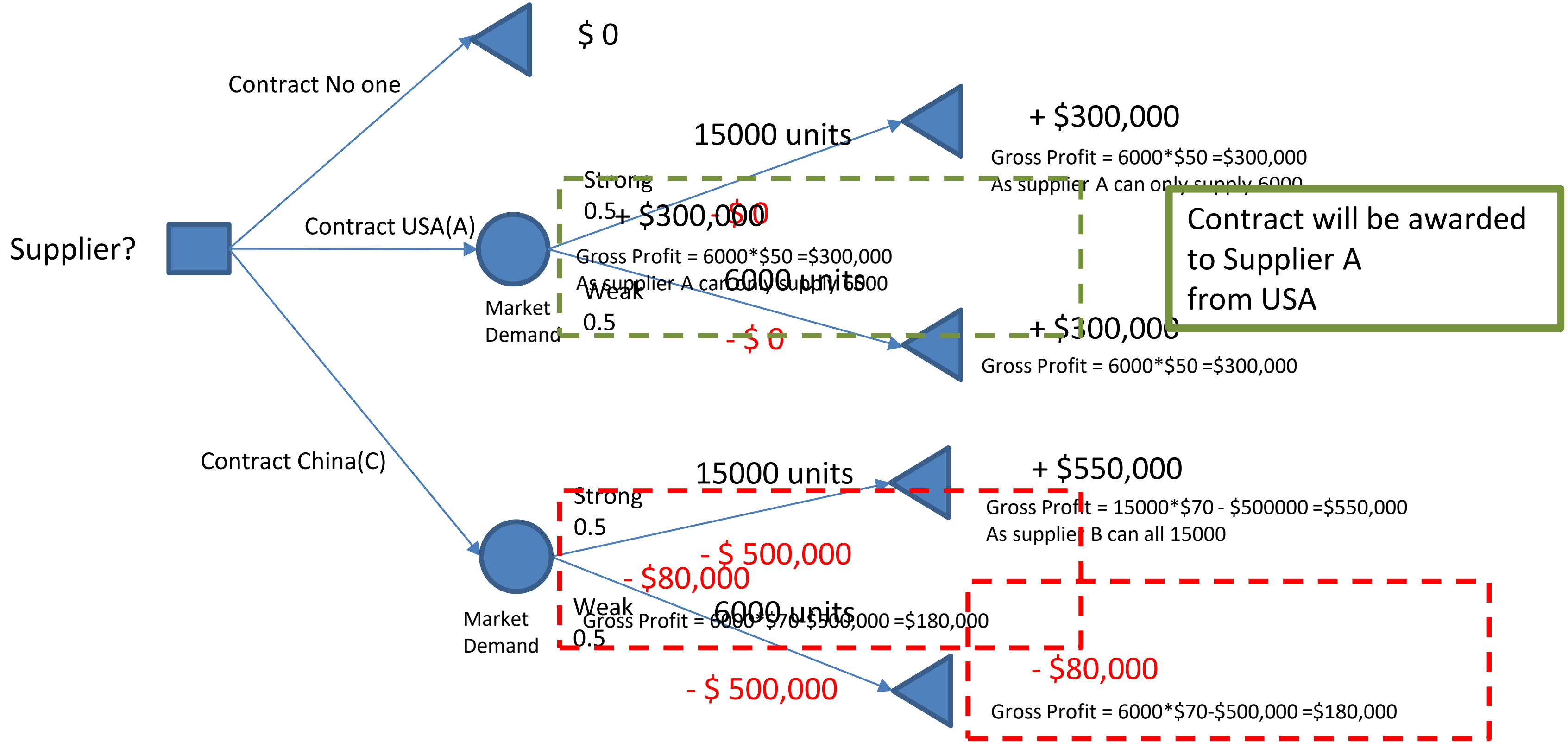
Common Decision Strategies

- Maxi – Mini Strategy : Risk Averse Strategy
  - Maximize the Minimum outcome
  - Avoids bad outcomes and ignores possibility of good outcomes with risk.

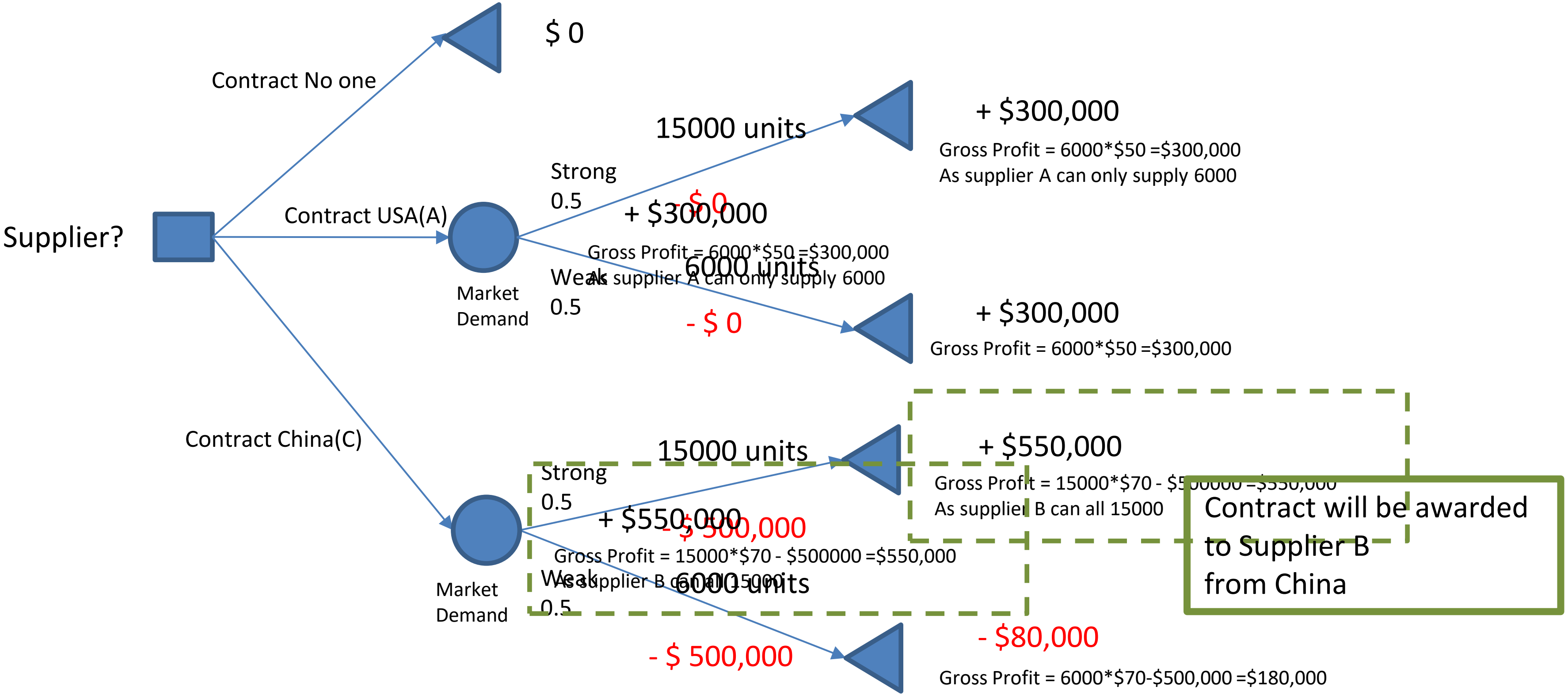
- Maxi – Maxi Strategy : Risk Seeking Strategy
  - Maximize the Maximum outcome
  - Seeks good outcomes and ignore possibility of bad outcomes

- Maximize Expected Value : Risk Neutral Strategy
  - Equal consideration of Good and Bad outcomes

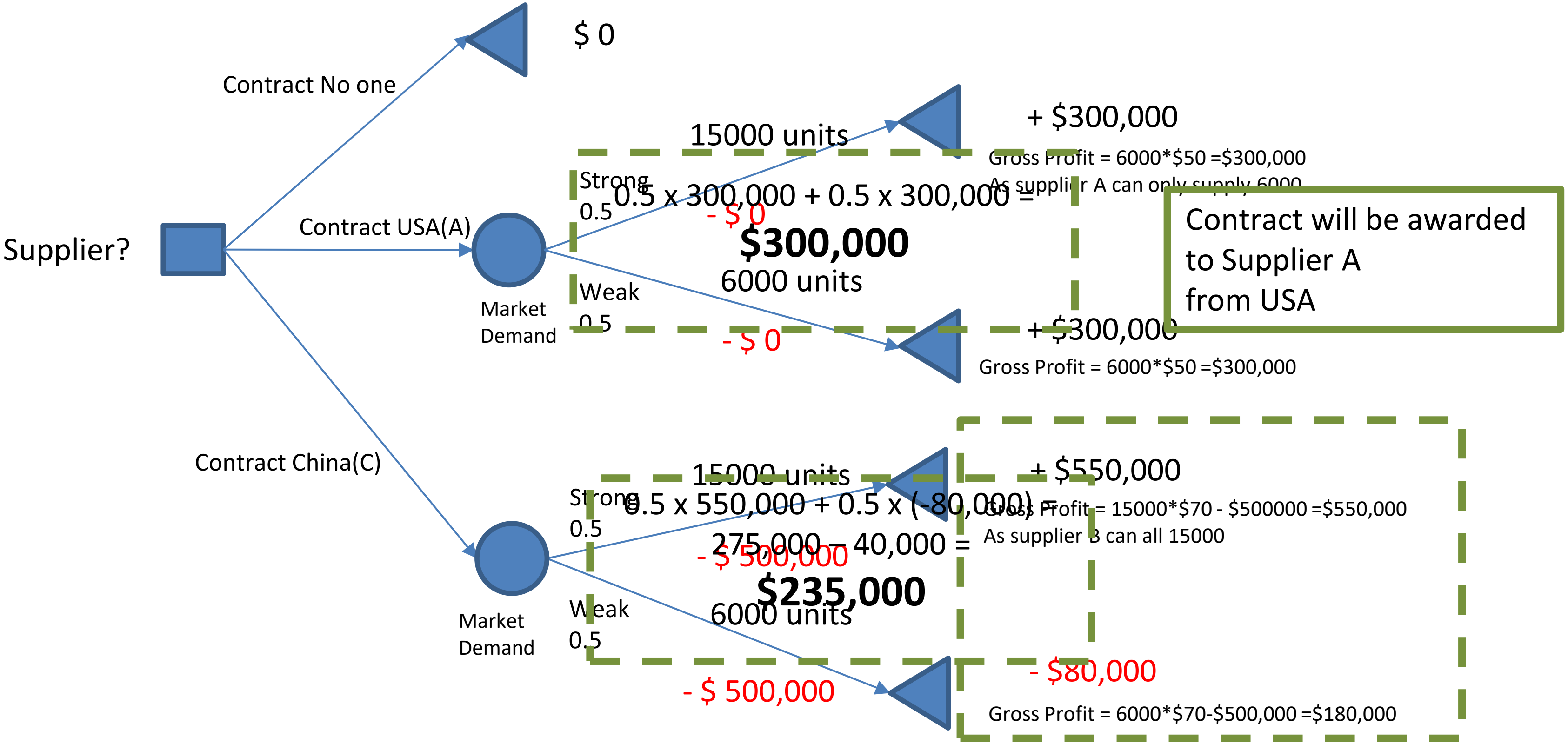
Maxi-Min Strategy



Maxi-Maxi Strategy



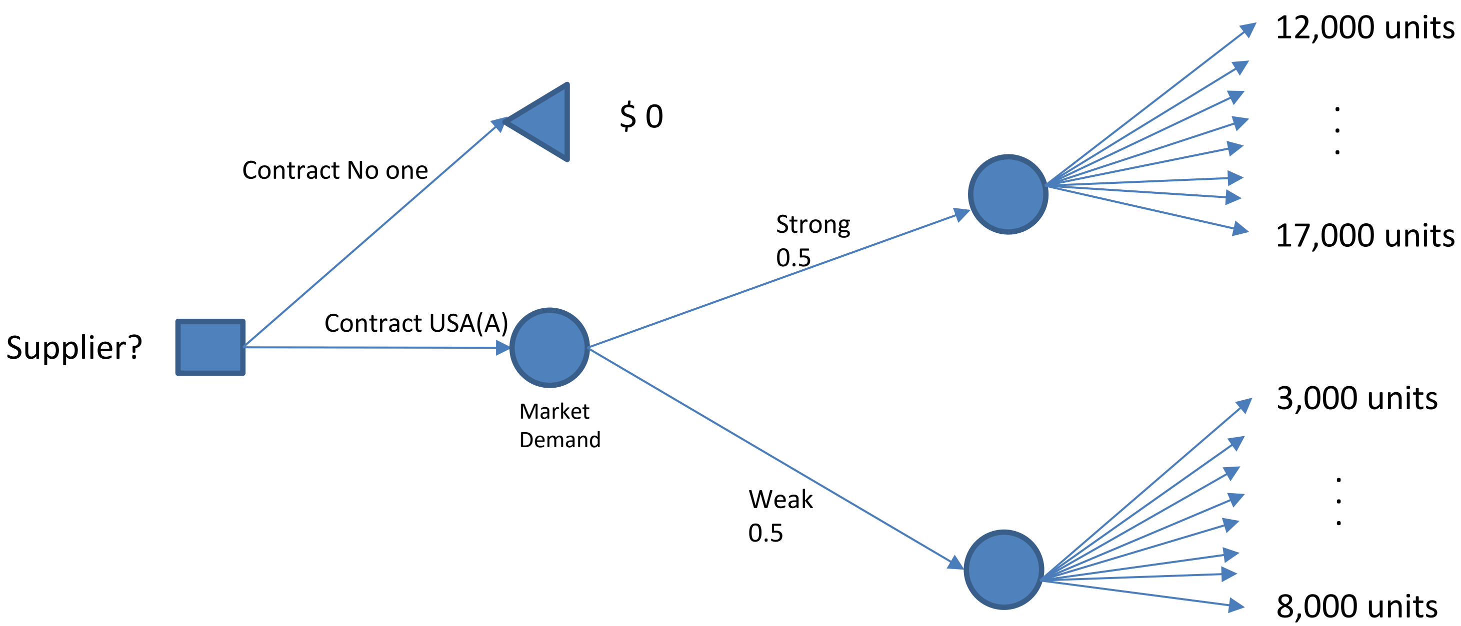
Maximize Expected Value Strategy



Datamites

What if the Demand when market in weak is not single value like 6000 units but a range like between 3000 to 8000 units?

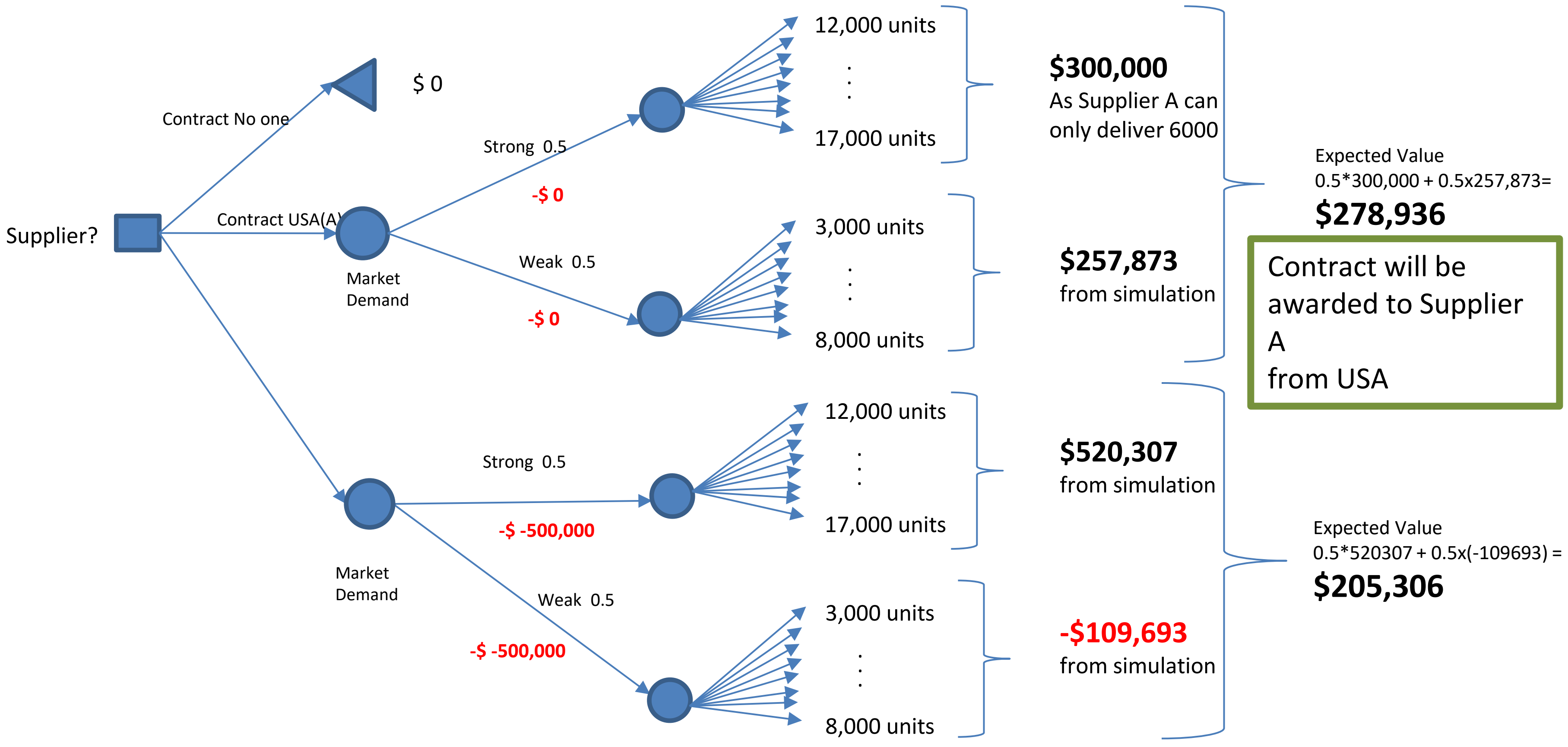
This is the usual scenario in practice.



As supplier A can only supply 6000 units and unit profit is $50

Profit = $50 * min { 6000, Demand}

It is difficult to model with decision tree branching; we use Monte Carlo simulation for this kind of decision process


Supplier?
Contract No one
$ 0
Contract USA(A)
Market Demand
Strong 0.5
-$ 0
Weak 0.5
-$ 0
12,000 units
17,000 units
$300,000
As Supplier A can only deliver 6000
3,000 units
8,000 units
$257,873
from simulation
Expected Value
0.5*300,000 + 0.5x257,873=
$278,936
Contract will be awarded to Supplier A from USA
Market Demand
Strong 0.5
-$ -500,000
Weak 0.5
-$ -500,000
12,000 units
17,000 units
$520,307
from simulation
3,000 units
8,000 units
-$109,693
from simulation
Expected Value
0.5*520307 + 0.5x(-109693) =
$205,306

What-If Analysis

What-If Analysis is a decision-making tool that helps you understand the **impact of changing input values** on the **outcome** of a model or formula.

In simple terms:

"What will happen to the result **if** I change this input?"

Types of What-If Analysis

1. Scenario Manager

Allows you to create and compare multiple input scenarios.

Example: Best Case, Worst Case, Most Likely Case.

2. Goal Seek

Works **backward**: You know the result you want and want to find the input needed.

Example: What interest rate is needed to achieve ₹10,000 return?

3. Data Table

Shows how changing one or two inputs affects the result.

Types: One-Variable and Two-Variable Data Tables.

Decision Trees

- Decision Trees are widely used in practice.
- Decision Tree can be built to more complexity with numerous decision points.
- Probabilities can be calculated from past data, expert judgement or Delphi method

Why BI for Data Analyst?

- Wide range of data visualization options to communicate insights effectively to non-technical stakeholders
- Making complex data more accessible and understandable.
- To create interactive dashboards that provide real-time insights into key performance indicators (KPIs)
- To explore and interact with the data, drill down into details, and customize their views, which enhances decision-making.
- Reduces manual reporting efforts.

Datamites

Why SQL for Data Analyst?

- SQL is a fundamental skill for data analysts as it provides the means to retrieve data from relational databases.
- Also to explore, analyze, clean, transform, summarize, and merge data stored in the databases.

Datamites

End of Module